When it Comes to Machine Preferences, Thurstone is the Model of Choice

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First version November 2024. This version August 6, 2026.

Abstract

Softmax output layers suggest, almost by construction, that large language models should obey Luce's Choice Axiom. They do not. In masked-token experiments a Thurstonian contest model predicted BERT's revealed preferences better in roughly 80% of cases. We extend the test to modern models with exact log-probability designs: across 461 permutation-controlled deletion cells, Thurstone beats Luce decisively (mean ΔKL $+0.033$, CI $[+0.020, +0.047]$), with the advantage concentrated wherever preference tuning has not already collapsed the distribution. Explicit menus behave differently: near-deterministic, no red-bus substitution, and occasional regularity violations (decoy effects) beyond any random utility model. Restriction on polysemous words connects these results to the stated-versus-generated gap of Cekinmez et al. (2026).

1 Introduction

Large language models can be interrogated cheaply and at scale, which makes the study of their revealed preferences attractive whether one regards them as incipient minds or as statistical mimics of our own predilections. Either way, a question from the psychology of choice arises immediately: when a model distributes probability over alternatives, what law governs how that distribution changes as the set of alternatives changes?

The workhorse answer in economics and psychology is Luce's Choice Axiom [5], which asserts that the probability of selecting item i from a set S is proportional to a fixed utility $u(i)$:

$$P(i \mid S) = \frac{u(i)}{\sum_{j \in S} u(j)}. \tag{1}$$

Its signature is Independence of Irrelevant Alternatives: the ratio $P(A)/P(B)$ is unchanged by the presence or absence of a third option C.

There is a natural alignment between this assumption and the token probabilities of language models, which invariably emerge from a softmax:

$$p_i = \frac{e^{z_i}}{\sum_{j=1}^{n} e^{z_j}}, \tag{2}$$

however complicated the transformer computation producing z_i may be. Delete item 1 from consideration—set $z_1 = -\infty$, a surgical procedure impossible in humans—and the remaining probabilities are the old ones renormalized by $1/(1 - p_1)$. In using softmax layers we are, in effect, heavily *suggesting* to the machine that it obey Luce's Choice Axiom.

It doesn't. This paper demonstrates that language models, despite the suggestion built into their output layer, exhibit preference behavior better modeled in the Thurstonian framework of noisy contests [7]. Section 2 states the two competing models. Sections 3–4 test them on masked language models. Section 5 extends the test to modern chat models using polysemous words, connecting the paradox to the stated-versus-generated gap recently documented by Cekinmez, Wu, Marjieh and Griffiths [8]. Section 6 draws out the implications: a mechanism for generative "mode collapse," a principled diversity knob, and a case for Thurstonian layers.

2 A choice of choice models

One might describe Equation (1) as an *urn model*: item k is represented by balls in an urn in proportion to $u(k)$, a choice is a random draw, and shrinking the choice set merely removes balls. The renormalization of probability under restriction then seems perfectly reasonable.

But perhaps preference is a horse race, not an urn. In the competing model each item i carries a location parameter a_i (a dislikability, say) and an associated performance $X_i \sim N(a_i, 1)$. Item i is chosen when its performance is the lowest draw. This is the model of Thurstone [7], and it makes different predictions about how choice probabilities respond when the choice set is altered: rather than removing a ball from the urn, we remove a horse from the race.

(As an aside, it *is* possible to engineer a contest that satisfies Luce's Choice Axiom, by taking $X_i \sim \mathrm{Exp}(h_i)$ with common location and differing hazard rates—or, by Yellott's theorem, with Gumbel noise [16]. The comparison here can therefore be viewed as between two noise families within the contest framework, which is why single-menu fits cannot separate them and the restriction, duplication, and context designs below are required.)

The practical obstacle to Thurstonian models has always been computation, not plausibility. A fast lattice algorithm for calibrating multi-entrant Thurstone models—recovering locations $\{a_i\}$ from winning probabilities and computing derived quantities for fields exceeding 100,000 entrants—was given in [1], and is what makes the experiments below routine.

3 Experimental design: masked language models

We probe the token probabilities of BERT [2] and RoBERTa [4], prompted to fill in blanks that invite choices among items of a category. Two designs are used.

In the first, the model answers an *original* question and a *qualified* question whose only difference is an adjective that shrinks the choice set:

1. *Original:* "My favorite state in the U.S. is [MASK] and I try to visit once a year."
2. *Qualified:* "My favorite Western state in the U.S. is [MASK] and I try to visit once a year."

In the second, we eliminate a single option explicitly, feeding the top answer of a first prompt back into a second:

1. *First:* "My favorite primary color is [MASK] because it is SOMETHING."
2. *Second:* "My two favorite primary colors are [ANSWER] and [MASK] because they are SOMETHING."

where SOMETHING ranges over many adjectives so that luck of phrasing is averaged out (the appendix of the working version lists roughly one hundred category/adjective combinations).

Both models make zero-parameter predictions of the restricted distribution from the unrestricted one. Luce's prediction is renormalization, Equation (1); in the two-step design this is the simplest Plackett–Luce [6] or Harville [3] construction. The Thurstonian prediction calibrates locations $\{a_i\}$ to the unrestricted token probabilities via $\mathrm{Prob}(X_i < X_k \ \forall k \neq i) = p_i$, removes the excluded contestant, and recomputes winning probabilities.

4 Results for masked language models

Table 1 shows the "favorite Western state" prompt for BERT. The Thurstonian column tracks the actual restricted probabilities more closely than Luce renormalization, which overweights the favorite (California) at the expense of the field. Table 2 repeats the exercise for "favorite girl's baby name," where the RMSE using Luce's probabilities is roughly 50% higher.

State	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
California	10.89	29.87	25.38	20.17
Arizona	6.74	18.50	17.20	10.00
Texas	5.50	15.10	14.59	7.68
Colorado	3.18	8.74	9.38	7.81
Oregon	2.89	7.92	8.67	8.57
Oklahoma	1.97	5.41	6.37	6.64
Nevada	1.66	4.55	5.54	10.18
Montana	1.59	4.37	5.36	14.27
Idaho	1.05	2.87	3.82	4.99
Wyoming	0.98	2.69	3.62	9.69

Table 1: Predicted probabilities under Luce's Choice Axiom and the Thurstonian model for the "favorite Western state" prompt (BertForMaskedLM). Both predictions use only the unqualified token probabilities; the Thurstonian model is closer to the actual qualified probabilities.

Across the full panel of qualified-question and single-item-elimination experiments, the Thurstonian model provides the more accurate prediction in approximately 80% of cases, with its advantage concentrated where the qualification is substantial and the entropy of the unrestricted distribution is low.

5 Polysemy as a preference probe

Cekinmez, Wu, Marjieh and Griffiths [8] recently turned an everyday property of language into a measurable distribution over meanings: give a model a bare polysemous word ("bolt") with no disambiguating context, sample many outputs, and classify which sense each expresses. Across 32 text and image generation models they report a striking ladder of normalized sense entropies: image generation 0.10, text generation 0.25, human subjects 0.47, and—when models are asked to *predict* the distribution of senses—roughly 0.73. Models know the ambiguity; they do not express it. The finding sits within a growing literature on generative homogenization [9] and

Name	Original (%)	Luce (%)	Thurstonian (%)	Actual (%)
Lily	1.56	20.35	16.18	14.52
Emily	1.15	15.06	13.31	16.29
Mia	0.86	11.25	11.02	13.18
Bella	0.85	11.12	10.93	8.84
Sarah	0.68	8.89	9.44	9.41
Rachel	0.59	7.66	8.56	7.15
Kate	0.59	7.63	8.54	8.44
Lauren	0.47	6.11	7.39	7.23
Brittany	0.46	6.04	7.33	7.55
Chloe	0.45	5.89	7.21	7.39

Table 2: The same comparison for "favorite girl's baby name." The RMSE using Luce's probabilities is approximately 50% higher than the Thurstonian RMSE.

its sharpening under preference tuning [10]; their own ablation shows DPO alone cutting sense entropy from 0.25 to 0.18.

Read through the lens of Section 2, the ladder is what a contest model produces. Let the stated distribution proxy the latent locations $\{a_i\}$; let each generation be a contest among noisy sense performances. When noise is small relative to the spread of locations, winning probability is a violently convex function of location differences, and a mildly uneven prior collapses onto its mode. One noise parameter then indexes the entire ladder, and the "multimodal gap" becomes a noise gap rather than a knowledge gap. The rival explanation is a power tilt: renormalized p_i^γ, the Luce family with a temperature, which is mechanically what RLHF sharpening, low-temperature decoding, and classifier-free guidance perform. The two families are distinguished not by the collapse itself but by behavior under choice-set restriction—precisely the instrument of Section 3, and an experiment run by neither paper until now.

5.1 Replication on a chat model

We replicated the design at pilot scale on Claude models (October 2025 generation): 15 polysemous words from the stimulus set of [8], 150 independent sentence generations per word, senses classified by a second model against the closed inventory, plus a stated distribution elicited with their "predicting itself" framing. The collapse replicates almost exactly: mean normalized sense entropy 0.252 generated (their text-model figure: 0.25) against 0.880 stated.

Fitting the two one-parameter families from stated to generated distributions gives a statistical tie (Thurstone $\sigma = 0.42$, RMSE 0.273; power tilt $\gamma = 2.25$, RMSE 0.273; Thurstone the better fit on 10 of 15 words), and the reason both fit poorly is itself informative: for several words the generated mode is not even the stated argmax ("port" generates *harbor* in every sample while rating *connector* most probable). Stated reports are a biased window on the latent locations, not merely a flattened one—consistent with [8], whose stated distributions overestimate even human diversity. Calibration should therefore use revealed behavior, which is what the restriction design does.

5.2 Choice-set restriction

For the six words whose unrestricted distributions were non-degenerate, we re-sampled with the modal sense excluded by instruction (150 samples per condition) and compared zero-parameter predictions of the restricted distribution: Luce renormalization against Thurstonian contestant removal, both calibrated to the add-half smoothed unrestricted sense counts.

Word	Luce RMSE	Thurstonian RMSE	Restricted distribution
bolt	0.450	**0.413**	fastener .68, run .18, lock .14
seal	**0.077**	0.089	animal .67, close .33
pitch	0.655	**0.630**	throw .06, tone .31, field .63
bat	0.559	**0.546**	animal 1.00, hit .00
match	0.758	**0.651**	firestick .82, contest .18
iron	**0.002**	0.037	metal .95, press .04, golf .01
pooled	0.495	**0.459**	

Table 3: Restriction test on polysemous words (Claude Haiku 4.5, 150 samples per condition, modal sense excluded by instruction). The Thurstonian prediction is closer pooled and on four of six words (bootstrap over words: $P(\text{Thurstone better}) = 0.90$).

The margin favors Thurstone, but the qualitative signature matters more than the margin: under restriction the model does not spread probability proportionally, as Luce requires. It *re-collapses* onto a new modal sense (match → *firestick* at .82; iron → *metal* at .95; bat → *animal* at 1.00). Remove the winning horse and the runner-up now wins nearly always—low-noise contest behavior, and the same low-entropy, heavy-qualification regime where the Thurstonian advantage concentrated in Section 4. Some rank flips under restriction exceed even the Thurstonian sharpening, and inventories with overlapping senses ("bat") contaminate the exclusion instruction; neither family survives those cells.

5.3 Exact probabilities on GPT models

Sampling noise can be removed entirely on models that expose token log probabilities. We ported the qualified-question design of Section 3 to GPT-4o-mini, GPT-4o and GPT-4.1: twenty category pairs ("my favorite color is" / "my favorite warm color is"), two phrasings, next-token distributions read exactly from the top-20 log probabilities, with the Thurstonian field restricted to a declared category inventory. Across 71 usable cells the comparison is a statistical tie with a Thurstonian lean: Thurstone wins 38 of 71 cells and has the lower pooled RMSE on all three models (e.g. 0.290 vs 0.295 on GPT-4o), but a bootstrap over cells does not separate the families ($p = 0.29$).

The instrument, however, has weakened for an interesting reason: the unqualified distributions of preference-tuned chat models are already near-degenerate ("the best metal is" puts 0.997 on *gold*), so most cells carry little discriminating information—under either model the surviving favorite keeps nearly everything, and the comparison hinges on tail probabilities. The masked models of Section 4 answered the unqualified question with entropy to spare (California at 10.9%); their preference-tuned descendants have already collapsed before the experimenter arrives. This is the same collapse measured by [8], now visible as a loss of experimental power:

alignment has pushed the interesting variation into the tails, where only exact log-probability designs can reach it.

5.4 Random elicitation restores the regime

The remedy is to ask for randomness rather than preference. "Name a random metal" followed by explicit single-item elimination—either an exclusion ("… that is not gold") or the two-slot continuation of Section 3 ("I already picked one random metal: gold. Name another…")—raises mean unqualified entropy to 0.38 and returns the experiment to the regime where the two families disagree. (Only partially: models are poor samplers even on request, with "a random planet" putting essentially all mass on *Mars*—the homogenization of [9] extends to explicit randomness.)

To control for phrasing and for which item is removed, the definitive battery deletes *each* of the top six observed items in turn, across 37 categories and three GPT models: 461 permutation-controlled cells, scored by the proper rule KL(actual ‖ prediction), with bootstrap confidence intervals over cells.

Stratum	n	mean ΔKL (Luce − Thurstone)	95% CI
all cells	461	$+0.033$	$[+0.020, +0.047]$
non-degenerate prior ($\tilde{H} > 0.2$)	323	$+0.044$	$[+0.028, +0.062]$
degenerate prior ($\tilde{H} \leq 0.2$)	138	$+0.007$	$[-0.012, +0.031]$

Table 4: Permutation-controlled deletion battery, exact log probabilities, KL scoring. Positive favors Thurstone. The Thurstonian advantage is decisive overall and concentrates precisely where the model retains genuine choice entropy.

Contest behavior governs wherever machine preference is genuinely stochastic; the Choice Axiom is competitive only in the degenerate cells that preference tuning has emptied of choice.

The largest effect of all comes from re-running the *original* stimulus set of this paper's 2024 version—99 categories with adjective-varied prompt pairs—in its original two-slot form ("My two favourite human organs are the [ANSWER] and the [MASK] because they are SOMETHING"), on GPT models with exact probabilities. Across 691 cells the conditional second choice decisively favors the contest model: mean ΔKL $+0.51$, 95% CI $[+0.43, +0.59]$, on every model and in both entropy strata. Predicting the runner-up given the winner is precisely where Luce renormalization fails hardest—the machine-preference analog of the Harville [3] overpricing of exactas that motivated the Thurstonian computation in the first place. The same questions asked of BERT in 2024 and of GPT-4-class models in 2026 return the same verdict, with a larger margin under exact measurement.

One further caution emerged with practical import for all such experiments: the "actual" restricted distribution depends materially on how the restriction is phrased. Holding everything else fixed, the exclusion and two-slot variants disagree about the restricted distribution by a median maximum gap of 0.14, and up to 0.65 (GPT-4o-mini names *Earth* after *Mars* with probability 0.94 under one phrasing and 0.56 under the other). Under surgical removal from a fixed preference state the two phrasings would coincide; in a language model the choice-set manipulation is itself a prompt intervention. Framing effects of this size sit above both choice models, and any account of machine preference that ignores them is incomplete.

5.5 Menus, duplicates, and context effects

Presenting the choice set explicitly—"pick one of these: ...," with menu order randomized and probabilities read from option-token log probabilities—changes the phenomenon entirely, in three ways.

First, menu choice by preference-tuned models is nearly deterministic. A transport menu (car, bus, train, bike, walk, taxi) under contextual conditioning behaves with perfect semantic sense—"it is raining" sends GPT-4o to *taxi* with probability 1.00, "there is a fuel shortage" to *bike* at 0.99—but the resulting distributions are so degenerate that menu-based deletion cannot separate the choice models (144 cells, ΔKL CI $[-0.028, +0.016]$), consistent with Table 4's stratification. The determinism extends to simulated agents: fictional election candidates with platform descriptions, evaluated through fictional voter personas, produce near-certain votes even when the persona is written to be genuinely cross-pressured, and candidate-withdrawal cells are consequently uninformative. The model represents a persona as an argmax decider; it does not carry within-persona randomness at all.

Second, machines are anti-red-bus. In Debreu's classic objection to the Choice Axiom [11], splitting a bus into a red bus and a blue bus should not double the bus vote; humans treat near-duplicates as substitutes, which correlated-noise (probit) models capture. Across ten duplicate-pair families the models show *no substitution at all*: the duplicated category's total share typically inflates, often beyond even the urn-doubling prediction (apple 0.08 $\rightarrow$ 0.53 when split into green and red), and a substitution baseline is the best predictor in only 8 of 25 cells against Luce's 17. (With exchangeable duplicates, Luce and independent Thurstone nearly coincide; the test's power is entirely against substitution.) Description salience—*red* apple is more vivid than apple—appears to dominate any shared-noise structure.

Third, where a menu does leave residual entropy, machine choice violates *regularity*—the property, shared by every random utility model including Thurstone's, that an added option cannot increase an incumbent's absolute share. Using two-attribute options (price versus quality) in randomized lettered menus, an asymmetrically dominated decoy [12] increased the target's absolute share in 8 of 18 family-model cells, and a compromise manipulation [13] in 4 of 18. The phone-plan family shows a textbook attraction effect on all three models (GPT-4o: target share 0.26 $\rightarrow$ 0.84 when its decoy appears). Machine preference, like human preference, is not a random utility model in the menu regime.

The menu regime admits a verdict stronger than any pairwise model comparison, because exact probabilities make it feasible to measure the *complete choice structure*: all eleven menus over a four-item universe, a measurement no human experiment delivers cleanly. By the Block–Marschak–Falmagne theorem [14, 15], such a structure is random-utility representable if and only if its Block–Marschak sums are nonnegative. Across six item families and three GPT models, **all 18 measured structures violate the conditions** (3–9 negative terms of 32 each; worst -0.58, far beyond measurement noise from order averaging). Fitting the structures anyway, the best possible random utility model (a distribution over all 24 rankings) leaves half the divergence unexplained (total KL 5.3, against 9.7 for the best Thurstone locations and 10.4 for the best Luce utilities—Thurstone the better parametric fit in 14 of 18 structures). In the menu regime, machine choice is not a random utility model at all.

Together the batteries locate the choice law in the *elicitation architecture*. Open-vocabulary recall behaves like a Thurstonian contest among retrieved candidates (Table 4); explicit menus behave like near-deterministic maximization perturbed by salience and context effects that no random utility model accommodates. Between them sits the sampled-sentence regime of [8], where the contest reading again does best.

6 Discussion

To the extent that LLMs mirror our own preferences, these results provide a rejection of Luce's Choice Axiom in favor of the Thurstonian model. The popularity of the former may owe much to computational convenience, a consideration the algorithm of [1] removes.

The polysemy results suggest something further: that generative "mode collapse" is not a defect of knowledge but the lawful output of a low-noise contest run over intact locations. That reframing carries a practical remedy. If choice is a contest, diversity is restored not by a power tilt—raising the sampling temperature, which distorts the tails and does not renormalize the way behavior actually renormalizes—but by inverting the contest to recover locations and resampling at a chosen noise level. The calibration is exactly the fast Thurstone computation of [1], and it offers a principled diversity knob without retraining.

Leaning into the irony of softmax machines disobeying Luce's Choice Axiom, the results also suggest that Thurstonian neural network layers might generalize better than softmax layers, the engineering challenge notwithstanding. Current architectures force hidden layers to fight the renormalization behavior their output layer implies; a contest-structured output layer would not need to.

Finally, this study is coarse-grained, and the pilot of Section 5 is small: fifteen words, one generation model, a single machine judge, sampling temperature not controlled. The full stimulus set and model panel of [8] would settle the one-parameter comparison with real power, and any model of machine decision-making that satisfies Luce's Choice Axiom might, on this evidence, be viewed with suspicion—and readily improved.

Acknowledgments

We thank the developers of BERT and RoBERTa for making their models accessible for research purposes.

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